

Deepam Minda

Applied ML Engineer | AI Engineer

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PROFESSIONAL SUMMARY

Applied ML Engineer with 4+ years of experience building and operating production AI systems across multimodal agents, document intelligence, computer vision, NLP, forecasting, and backend automation. Owns systems end to end - from data and model evaluation through APIs, orchestration, cloud deployment, monitoring, and human-in-the-loop operations - with measurable improvements in automation coverage, throughput, and decision quality.

TECHNICAL SKILLS

Generative AI and Agents: LLM agents, multi-agent architectures, LangChain, LangGraph, DSPy, RAG, tool calling, structured outputs, persistent memory, prompt and harness engineering, multimodal AI, evaluation, OpenAI, Gemini, OpenRouter, Pydantic

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, SAM, DenseNet, CLIP, sentence transformers, SetFit, fine-tuning, few-shot learning, contrastive learning, classification, segmentation, forecasting, model evaluation

Backend and MLOps: Python, FastAPI, Celery, REST APIs, Docker, AWS EC2/ECS/SQS/SNS, GCP Cloud Functions, GitHub Actions, CI/CD, event-driven and scheduled workflows, authentication, logging, monitoring, Sentry, CloudWatch

Data: SQL, PostgreSQL, MongoDB, Redis, MySQL, Databricks, pandas, NumPy, SQLAlchemy, data pipelines, feature engineering

PROFESSIONAL EXPERIENCE

Farmart

Data Scientist I

Associate Data Scientist

May 2023 - Present

Oct 2024 - Present

May 2023 - Oct 2024

- Architected and shipped a modular document AI review platform processing approximately **500 documents/day**, combining multimodal classification and extraction, configurable business rules, ML validators, and human-in-the-loop checkpoints.
- Raised automated coverage from **0% to 60%+** on the primary document pipeline, reaching **70% in active flows**, and achieved approximately **30% automation** on a second document type while retaining reviewable decisions and exception handling.
- Designed production multimodal agent workflows across product and messaging channels, supporting text, image, document, and voice inputs with retrieval-augmented context, streaming responses, tool calling, and authenticated downstream actions.
- Engineered the agent execution layer for query routing, multi-agent orchestration, chat and session state, persistent memory, media ingestion, tool invocation, and integration with operational APIs.
- Built a retrieval-augmented content automation pipeline spanning source discovery, ingestion, relevance scoring, LLM metadata tagging, DSPy-driven draft generation, AI-assisted editing, and human review; increased peak throughput from **6-8 to 14-15 articles/day**.

Farmart (continued)

May 2023 - Present

- Developed computer vision systems for quality assessment across **2 commodity categories** using SAM segmentation, DenseNet classifiers, and contrastive representation learning; achieved **85%+ sample-weighted accuracy** and reduced buyer-retailer deduction disputes by **14%**.
- Built label-efficient image and text classification workflows using CLIP, sentence transformers, contrastive learning, and SetFit-style fine-tuning; reached approximately **80% accuracy** in selected moderation and compliance use cases with as few as **100 labels per class**.
- Developed commodity price forecasting experiments over sparse multi-source market data using web scraping, feature engineering, linear regression, random forests, and gradient-boosted trees; achieved **1.3% MAPE for 7-day** and **2.8% MAPE for 30-day** evaluation horizons.
- Built and maintained internal AI and data tooling with Python and FastAPI, Celery/SQS worker pipelines, scheduled jobs, review queues, and dashboards across AWS and GCP, backed by PostgreSQL, MongoDB, Redis, MySQL, and Databricks with CI/CD, logging, and monitoring.

Farmart

Feb 2023 - May 2023

Data Science Intern

- Fine-tuned and deployed a visual question answering model for multi-field document extraction, owning dataset creation, model training, API deployment, and post-launch monitoring from initial prototype through the first production version.

Scaler

Dec 2022 - Feb 2023

Product Analyst Intern

- Reviewed and developed data science, machine learning, and deep learning educational content and researched topics for new technical articles.

Doubtnut

May 2022 - Nov 2022

Machine Learning Engineer Intern

- Trained and deployed a deep learning classifier for Hinglish profanity detection in user-generated content, achieving **90%+ accuracy** and **0.84 F1** on offline evaluation.
- Implemented Whisper-based transcription and translation pipelines for short- and long-form video content, and prototyped a geospatial clustering-based recommendation system.

EDUCATION

Delhi Technological University, Delhi

2019 - 2023

B.Tech. in Electrical Engineering | GPA: 7.8/10

LEADERSHIP

Unmanned Ground Vehicles, DTU | Corporate Head

2021 - 2023

Led corporate affairs across a 30-member student engineering team and raised INR 400,000 for its IGVC vehicle project.